Develop a Java program to create a class Book with members Book ID, Title, Author, and Price. Use a constructor to initialize the details. Include methods to display book information, overload search() to search by Book ID and Title, and create a method that accepts another Book object and returns the costlier book. Use a static member to maintain the total number of books created. In the main class, create multiple book objects and demonstrate all the operations.

Code:

```
class Book {
   private int bookid;
   private String title;
   private String author;
   private double price;

   private static int totalBooks = 0;

   public Book(int bookid, String title, String author, double
price) {
      this.bookid = bookid;
      this.title = title;
      this.author = author;
      this.price = price;
      totalBooks++;
   }
   public int getBookid() {
      return bookid;
   }
   public String getTitle() {
      return title;
   }
   public void displayBookInfo() {
      System.out.println("Book ID: " + bookid);
      System.out.println("Title: " + title);
      System.out.println("Author: " + author);
      System.out.println("Price: $" + price);
      System.out.println("------------------");
   }
   public static int getTotalBooks() {
      return totalBooks;
   }

   public static boolean search(Book[] books, int id) {
      for (Book book : books) {
         if (book.getBookid() == id) {
            System.out.println("Book found with ID " + id +
":");
            book.displayBookInfo();
```

```
            return true;
         }
      }
      System.out.println("Book with ID '" + id + "' not
found.");
      return false;
   }

   public static boolean search(Book[] books, String title) {
      for (Book book : books) {
         if (book.getTitle().equalsIgnoreCase(title)) {
            System.out.println("Book found with Title '" +
title + "':");
            book.displayBookInfo();
            return true;
         }
      }
      System.out.println("Book with Title '" + title + "' not
found.");
      return false;
   }

   public Book getCostlierBook(Book otherBook) {
      if (this.price >= otherBook.price)
         return this;
      else {
         return otherBook;
      }
   }
}

public class BookDetails {
    public static void main(String[] args) {
        Book b1 = new Book(101, "Java 1", "Carl", 45.99);
        Book b2 = new Book(102, "Big Nate", "Tom", 52.50);
        Book b3 = new Book(103, "Harry Poter", "Watson",
39.95);

        Book[] inventory = {b1, b2, b3};

        System.out.println("Total books created: " +
Book.getTotalBooks());

        System.out.println("--- All Books ---");
        for (Book b : inventory) {
            b.displayBookInfo();
        }
```

```
        System.out.println("--- Search by ID ---");
        Book.search(inventory, 102);

        System.out.println("--- Search by Title ---");
        Book.search(inventory, "Java 1");

        System.out.println("--- Costlier Book Comparison ---
");
        Book costlier = b1.getCostlierBook(b2);
        System.out.println("Between '" + b1.getTitle() + "'
and '" + b2.getTitle() + "', the costlier book is: ");
        costlier.displayBookInfo();
    }
}
```

Output:

```
 ----jGRASP exec: java -Xint -Xrunjdwp:transport=dt_socket,suspend=y,server=y BookDetails
 ----jGRASP: Connected to debugger.
Total books created: 3
--- All Books ---
Book ID: 101
Title: Java 1
Author: Carl
Price: $45.99
------------------
Book ID: 102
Title: Big Nate
Author: Tom
Price: $52.5
------------------
Book ID: 103
Title: Harry Poter
Author: Watson
Price: $39.95
------------------
--- Search by ID ---
Book found with ID 102:
Book ID: 102
Title: Big Nate
Author: Tom
Price: $52.5
------------------
--- Search by Title ---
Book found with Title 'Java 1':
Book ID: 101
Title: Java 1
Author: Carl
Price: $45.99
------------------
--- Costlier Book Comparison ---
Between 'Java 1' and 'Big Nate', the costlier book is:
Book ID: 102
Title: Big Nate
Author: Tom
Price: $52.5
------------------

 ----jGRASP debugger: run in canvas ended, debugger will hold.
 ----jGRASP: Process ended by user.
```